

About Validation Method Standard

Governing How Validation Is Performed

Knowing what is being validated is not enough.

Knowing why it is being validated is not enough.

Defining explicit Validation Criteria is not enough.

A validation process must also establish **how those criteria can legitimately be examined**.

Validation Method Standard (VMTS) is the fourth Parent Standard of **VALIDOS® — Validation Governance Architecture**, governing the methodological conditions under which Validation Criteria are translated into controlled, justified, appropriate, and validation-ready methods.

VMTS addresses one fundamental question:

Which validation method, or combination of methods, can legitimately and sufficiently examine the established Validation Criteria under the governed validation conditions?

Why Validation Method Standard Exists

Validation results depend not only on what is evaluated, but also on how the evaluation is performed.

The same Validation Object may produce materially different conclusions when examined through different:

- methodologies,
- assumptions,
- configurations,
- samples,
- benchmarks,
- scenarios,
- instruments,
- analytical techniques,
- or evaluation procedures.

A method may be scientifically recognized but inappropriate for a particular Validation Object.

A technically sophisticated method may fail to examine the criterion that actually matters.

Several validation methods may appear independent while relying upon the same underlying assumptions or evidence.

A method may also be changed after results become visible, creating methodological bias.

VMTS exists to make these methodological choices explicit and governable.

Method After Criteria

VALIDOS® establishes a deliberate methodological sequence:

**Validation Object → Validation Purpose & Scope → Validation Criteria
→ Validation Method**

The method does not define what validity means.

That responsibility belongs to the preceding Validation Criteria governance.

VMTS receives an established Validation Criteria Framework and determines how those criteria can legitimately be examined.

This preserves a critical separation:

Criteria define what must be evaluated.

Methods define how it will be evaluated.

Without this separation, validation methodology can begin determining the conditions it was originally supposed to test.

Method Selection Must Be Justified

A validation method should not be selected merely because it is:

- familiar,
- inexpensive,
- available,
- automated,
- widely used,
- technically advanced,
- historically preferred,
- or likely to produce a favorable result.

The selected method must be capable of examining the relevant Validation Criterion under the applicable validation conditions.

VMTS therefore treats method selection as a governance decision requiring methodological justification.

There Is No Universal Validation Method

Different Validation Objects require different forms of examination.

A factual claim may require documentary and source-based validation.

A machine-learning model may require statistical, empirical, adversarial, and operational validation.

A simulation may require model validation, boundary validation, scenario validation, and comparison against observed reality.

A physical system may require measurement and testing.

A decision may require validation of inputs, reasoning conditions, authority, evidence, and contextual applicability.

VMTS does not attempt to replace these methods with one universal procedure.

Instead, it provides the governance architecture through which appropriate methods can be selected and controlled.

Method-to-Criterion Alignment

Every material Validation Criterion should have an identifiable methodological path.

The governing relationship is:

Validation Criterion → Methodological Requirement → Validation Method

A validation activity does not automatically provide coverage simply because something was tested.

The method must actually be capable of examining the condition represented by the criterion.

This prevents **validation activity** from being mistaken for **validation coverage**.

Method-to-Object Alignment

The method must also fit the Validation Object.

A methodology validated for one:

- object type,
- population,
- system configuration,
- environment,
- domain,
- or operational condition

cannot automatically be transferred to another.

VMTS therefore requires the relationship between the method and the actual Validation Object to remain explicit.

Assumptions Must Remain Visible

Validation methods frequently depend upon assumptions.

These assumptions may concern:

- data,
- independence,
- sampling,
- environmental stability,
- system behavior,
- population characteristics,
- causal relationships,
- measurement reliability,
- or expected operating conditions.

An apparently rigorous validation method can become unreliable when a material assumption is false.

VMTS therefore treats methodological assumptions as governed conditions rather than invisible background expectations.

Limitations Are Part of Validation

Every method has limits.

A method may work well within one operating range but poorly outside it.

It may detect common failures but not rare failures.

It may establish correlation but not causation.

It may represent controlled environments but not real-world conditions.

It may evaluate average performance while concealing important edge cases.

VMTS requires material limitations to remain visible throughout the validation lifecycle.

A limitation is not automatically a methodological failure.

An undisclosed limitation can become one.

Multi-Method Validation

Complex Validation Objects may require more than one method.

VALIDOS® therefore allows validation to use multiple methodological approaches.

Methods may operate:

- sequentially,
- simultaneously,
- independently,
- complementarily,
- comparatively,
- adversarially,
- or conditionally.

VMTS governs how those methods relate to one another.

Simply increasing the number of methods does not necessarily increase validation strength.

Methodological Diversity

Several methods may appear different while sharing the same:

- dataset,
- model,
- assumptions,
- benchmark,
- evaluator,
- measurement mechanism,
- or analytical dependency.

This creates the possibility of **false methodological diversity**.

VMTS therefore distinguishes between:

multiple methods

and:

meaningfully independent methodological perspectives.

Where validation confidence depends upon independent confirmation, genuine methodological independence becomes important.

Methodological Triangulation

Where appropriate, different methods may examine the same Validation Criterion from independent or complementary perspectives.

This can create methodological triangulation.

For example:

Empirical Observation + Statistical Analysis + Independent Expert Assessment

may provide a stronger basis than repeatedly applying one analytical method.

However, triangulation is valuable only when the methods contribute genuinely distinct validation information.

Methodological Conflict

Different methods may produce different results.

This does not automatically mean that one method is wrong.

The difference may reveal:

- different assumptions,
- different sensitivities,
- different populations,
- different evidence,
- different environmental conditions,
- different methodological boundaries,
- or genuine uncertainty.

VMTS requires material methodological disagreement to remain visible rather than being silently resolved in favor of the preferred result.

Configuration Matters

A method name alone does not fully describe a validation method.

For example, saying that a system was evaluated through "statistical validation" does not establish:

- the sample,
- parameters,
- confidence conditions,
- population,
- test window,
- statistical assumptions,
- thresholds,
- or evaluation configuration.

VMTS therefore governs material methodological configuration.

This creates a more precise methodological record of **how validation**

was actually performed.

Calibration Matters

Some methods require calibration before their outputs can be meaningfully interpreted.

Calibration may rely upon:

- known reference conditions,
- control samples,
- validated instruments,
- baseline systems,
- benchmark environments,
- or other reference states.

VMTS ensures that calibration remains identifiable and distinguishable from the validation result itself.

Method Readiness Before Execution

A method should not enter validation execution simply because it has been selected.

Before execution, the Validation Method Framework should establish that:

- methods are suitable,
- relevant criteria are methodologically covered,
- configurations are defined,
- assumptions are visible,
- limitations are documented,
- required calibration exists,
- independence requirements are addressed,
- and material methodological conflicts have been governed.

This creates a methodological readiness gate between design and execution.

Changing the Method Changes the Validation

A material change to a validation method may alter the meaning of the resulting validation.

Changes may affect:

- comparability,
 - evidence,
 - sensitivity,
 - confidence,
-

- interpretation,
- or final determination.

VMTS therefore requires material method changes to remain versioned and traceable.

Historical validation results must remain connected to the method under which they were produced.

Method Is Not Evidence

VALIDOS® maintains another important separation:

Method ≠ Evidence

A method determines how a condition is examined.

Evidence is the information used or produced through that examination.

A strong method cannot compensate for fundamentally unsuitable evidence.

Likewise, strong evidence can be misused through an inappropriate method.

These responsibilities therefore remain separate governance spaces within VALIDOS®.

Method Is Not Execution

VMTS designs and governs the methodological structure.

It does not govern whether the method was actually executed correctly in a specific validation event.

That responsibility belongs to subsequent Validation Execution governance.

This distinction enables VALIDOS® to separately ask:

Was the method appropriate?

and:

Was the approved method actually followed?

Both questions matter.

Position Within VALIDOS®

Validation Method Standard is the fourth Parent Standard of **VALIDOS®**

— Validation Governance Architecture.

Its architectural position is:

**Validation Object → Validation Purpose & Scope → Validation Criteria
→ Validation Method → Validation Evidence Fitness → Validation
Source Reliability → Validation Execution → Validation Determination
→ Validation State → Validation Reliance & Revalidation**

The first Parent Standard establishes:

What exactly is being validated?

The second establishes:

Why is it being validated and within what boundaries?

The third establishes:

Against which conditions will validity be judged?

The fourth establishes:

How can those conditions legitimately be examined?

This creates the methodological transition from validation definition into validation operation.

Relationship to Other OOF® Architectures

VMTS operates alongside complementary OOF® governance architectures.

GOA™ may provide governance conditions affecting methodological authority and control.

INTEGROS® may provide integrity conditions requiring validation.

ORA™ may provide operational-reality conditions against which methods must remain applicable.

OBIDENTITY® may provide identity, provenance, ownership, and continuity information relevant to methodological objects and dependencies.

AGA™ may provide accountability conditions for methodological responsibility.

RIS™ may influence required methodological depth according to risk.

AIG®, CLIA®, MGIATM, and ASGATM may provide specialized governance conditions requiring appropriate validation methods.

VMTS does not replace scientific, engineering, statistical, technical, audit, or domain-specific methodologies.

It provides the governance structure through which those methodologies can become controlled components of a validation process.

Why It Matters

Without governed validation methodology, two organizations may claim to have validated the same type of system while performing fundamentally different examinations.

VMTS makes it possible to ask:

- which method was used,
- why it was selected,
- which criterion it examined,
- whether it was appropriate for the Validation Object,
- which assumptions it depended upon,
- which limitations applied,
- how it was configured,
- whether it was independently corroborated,
- which version was used,
- and whether the method was sufficiently ready before execution.

This transforms validation methodology from an implicit technical choice into an explicit governance object.

Foundational Principle

A validation method is legitimate only when it is fit for the criterion it is intended to examine.

Canonical Closing Statement

Validation Method Standard establishes the methodological-design layer of VALIDOS® — Validation Governance Architecture. By governing how validation methods are selected, justified, aligned, configured, calibrated, combined, constrained, versioned, and prepared for execution, VMTS ensures that established Validation Criteria are examined through methods appropriate to the Validation Object and governed validation conditions. VALIDOS® therefore separates the question of what must be demonstrated from the question of how it should be demonstrated, preserving methodological clarity, traceability, and legitimacy throughout the validation lifecycle.

Parent Resources

→ [VALIDOS® Architecture Map](#)

→ [VALIDOS® Complete Standards & Modules Index](#)

Related Documents

→ [Validation Method Standard](#)

→ [VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

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